

Reliability Challenges and Solutions for Large Load Integration in Bulk Power Systems

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Abstract—The increasing number of Large Loads, such as data centers and cryptocurrency miners, is introducing new reliability challenges for Bulk Power Systems (BPS). As seen in recent industry events, these loads have become significant contributors to customer-initiated load reduction, oscillations, and frequency transients. While these Large Loads form the basis for our digital economy and support modern infrastructure, it is important that they do not adversely impact the BPS reliability. This paper investigates various BPS events that primarily involve Large Loads and develops a taxonomy of root-causes related to equipment design, control systems, and the software operating these loads. This taxonomy is utilized to guide further research into solutions for these issues, and the paper proposes both facility-level and grid-level mitigation techniques to address the identified challenges.

Index Terms—Data centers, Data center power, Cryptocurrency, Power system protection, Power system reliability, Power system stability

I. INTRODUCTION

The electric power industry is seeing an increase in demand, with data centers and cryptocurrency miners being key drivers in load growth [1]. These loads are primarily classified as “Large Loads”, i.e., loads that are larger than loads historically connected and can threaten the security of the bulk power system (BPS). The NERC Large Load Task Force defines these loads as “any commercial or industrial individual load facility or aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS due to its demand, operational characteristics, or other factors” [2]. There have been numerous reported events where grid disturbances, such as transmission faults, led to customer-initiated load reduction [3]–[5]. During these events, the voltage sensitivity of the computing and power distribution equipment played a key role in the initiation of customer load reduction. These Large Loads have also been the source of oscillations that were detected from the BPS [6], [7]. To ensure BPS reliability, it is extremely important that system operators understand the reliability risks that Large Loads pose so they can be effectively mitigated.

The challenges and risks posed by Large Loads to the BPS are not sufficiently documented and discussed in the existing literature. This paper fills the gap by identifying and characterizing these risks and challenges in detail. A comprehensive evaluation of various real-life industry incidents involving

Large Loads is presented. These include (a) impacts on Large Loads due to incidents in BPS and (b) impact on the BPS due to incidents involving Large Loads. Further, a taxonomy is introduced that classifies Large Load risks by root causes such as equipment design, control strategies, and operational characteristics. Finally, mitigation and design solutions for issues related to Large Load integration from both (a) facility and (b) grid side are systematically classified according to their scope and applicability.

The rest of the paper is organized as follows: Section II provides a background of important characteristics for Large Loads relevant to the BPS risks. Section III covers the general risks Large Loads pose to the BPS and disturbances involving Large Loads. Section IV covers Large Load facility-side mitigation and solutions to these risks, followed by grid-side mitigations and solutions in Section V. Finally, the conclusions are presented in Section VI.

II. BACKGROUND ON LARGE LOADS

The NERC Large Load Task Force [2] cites data centers, cryptocurrency mining facilities, hydrogen electrolyzers, manufacturing plants, and arc furnaces as examples of Large Loads. In this paper, we focus on data centers and cryptocurrency mining facilities. Data centers support much of our modern digital infrastructure, providing compute and storage for end-users. In recent years, data centers focusing on artificial intelligence (AI) model training and inference are increasingly being built. Compared to traditional data centers, these AI data centers generally have a much higher load demand and a highly variable load pattern [3], [8]. Critically, data centers have (a) a power distribution system including power supplies to convert AC power to DC, and (b) uninterruptible power supply (UPS) units that provide short-term battery backup capabilities. The power supplies and UPS units are typically designed to conform to the Information Technology Industry Council (ITIC) curve or a ride-through curve, which specifies the voltage limits of computer equipment across time.

In addition to data centers, cryptocurrency miners are also driving load growth. Cryptocurrencies such as Bitcoin are decentralized currencies on a blockchain where a peer-to-peer network maintains a record of transactions, and new coins can be unlocked through cryptocurrency mining. These miners use specialized computing equipment made with application

TABLE I
IMPACT ON LARGE-LOADS DUE TO POWER SYSTEM FAULTS

ID	Entity	Fault Description	Impact on Large Load	MW Load Affected	Remarks
1	AEP	Normally cleared 138 kV phase-to-ground faults	One Data Center rides through 1p fault, 1st, 2nd, and 3rd reclose attempt. Another Data Center trips after 2nd reclose attempt. Other data centers ride through. Crypto-mine rides through.	68 MW	Protection logic set to trip after 3 voltage dips occurring within 1 minute. Both data centers and crypto-mines are constant power loads during the voltage dips.
		Normally cleared 138 kV phase-to-ground faults	One Data Center rides through initial 1p fault, 1st reclose attempt but trips after 2nd reclose attempt. Other data centers ride through. Crypto-mine rode through.	80 MW	
2	Dominion	230 kV Line locked out during a storm event. A total of 6 reclosing attempts with 3 at each substation	Initial load reduction after first and second reclosing shots. Most load reduction was after third reclosing shot. Some data centers rode through albeit with a brief change in load.	1470 MW	Protection logic set to trip when a certain number of voltage disturbances are seen within a certain time. Need high-resolution monitoring data to review facilities performance.
		230 kV transmission line fault three auto-reclose attempts at each end of the line		1800 MW	
3	EirGrid	Phase-phase fault on 220 kV line - tripped	Data centers disconnected from the grid	80 MW	UPS voltage protection settings set to 10% from nominal - data centers disconnect within ms of faults. Faults were cleared in 61 ms, then 80 ms/90 ms (reclose trip) respectively.
		220 kV line tripped, reclosed and tripped for a single phase to ground fault		204 MW	
		220 kV reactor fault		321 MW	
4	ERCOT	multiple events single phase and 3 phase faults	Large Load reduction as single phase voltage dipped below 0.7 p.u. Large Loads reduction below a voltage of 0.75 p.u. for a 3LG fault on 345 kV	107 MW - 432 MW	Need for validated dynamic models and high resolution monitoring.

TABLE II
IMPACT ON POWER SYSTEM DUE TO LARGE-LOADS

ID	Entity	Event / Issue	Impact on Power System
1	ERCOT	Older firmware issues	~25 MW peak to peak and with ~23 Hz oscillations
2	EirGrid	Regular, cyclical fluctuations in data center demand	~10 mHz-50 mHz frequency transients; drop of 5-15% of nominal steady-state system frequency; system oscillations; interaction with governors and active power control from wind/solar
2	Dominion	Unstable UPS in the data center	14.7-14.8 Hz oscillations

specific integrated circuits to mine Bitcoins. Miners typically display a constant load pattern running at full consumption unless triggered to reduce demand from external factors. Experimental testing of cryptocurrency miners such as S19 Pros has validated that their ride-through curves are similar to the ITIC curve [9], [10].

From a power system reliability perspective, it is the interaction of the power conversion infrastructure, including power supplies and UPS units at the Large Load facility, with the BPS that poses one of the most urgent concerns, which we seek to investigate in this paper.

III. LARGE LOAD THREATS TO BULK POWER SYSTEM

The rapid growth and operations of Large Load facilities pose significant threats to the BPS, especially when there is unplanned load reduction in response to transmission faults or voltage disturbances. Sudden load reduction from the BPS perspective can lead to frequency overshoot and trigger over-frequency protection [11]. Oscillations arising from the operations of these Large Loads and their associated controls can also have widespread impacts on power system stability. This section provides an overview of recent large-load events and their root causes.

A. Fault-Induced Customer Initiated Load Reduction/Tripping

Customer-initiated load reductions have been observed at Large Loads, including data centers and cryptocurrency facilities, following various normally cleared single-phase or phase-to-phase transmission fault events. In these facilities, protection schemes are configured to (a) trip after experiencing

multiple voltage disturbances in rapid succession, or (b) the equipment trips following voltages outside of its ride-through capabilities. Some recent examples include:

1) *Dominion Energy*: In 2024, a storm led to a 230-kV transmission permanent line fault that was caused due to the failure of a lightning arrestor, which ultimately locked out the line. There were three auto-reclosing actions performed at each end of the line over 85 seconds. After three reclosing attempts, the load began to reduce demand and transfer to backup generation. It was found that numerous data centers had a “3-strike” rule where the protection schemes count the number of voltage disturbances, and upon reaching three or more strikes, they switch to backup generation or trip. A total of 32 substations were impacted, and the area control error (ACE) was increased by 1470 MW [3]. In a similar incident that occurred in 2025, loads from multiple data centers were transferred off the grid during a protection reclosing event, leading to an increase in ACE by 1800 MW [12].

2) *ERCOT*: In the ERCOT Interconnection, there have been numerous events over the past two years where a transmission fault caused reductions in power electronic load demand as the Large Loads did not ride through the resulting voltage disturbance. It was noted that the single line to ground faults caused significant reductions for shallow positive sequence voltage dips. In these cases, when the faulted phase was reduced to below 0.7 p.u., load reduction was seen, and most load reduction was only in the faulted phase. Some loads were noted to be more sensitive to the voltage disturbances, as there was a large variance in the load behavior in response to the

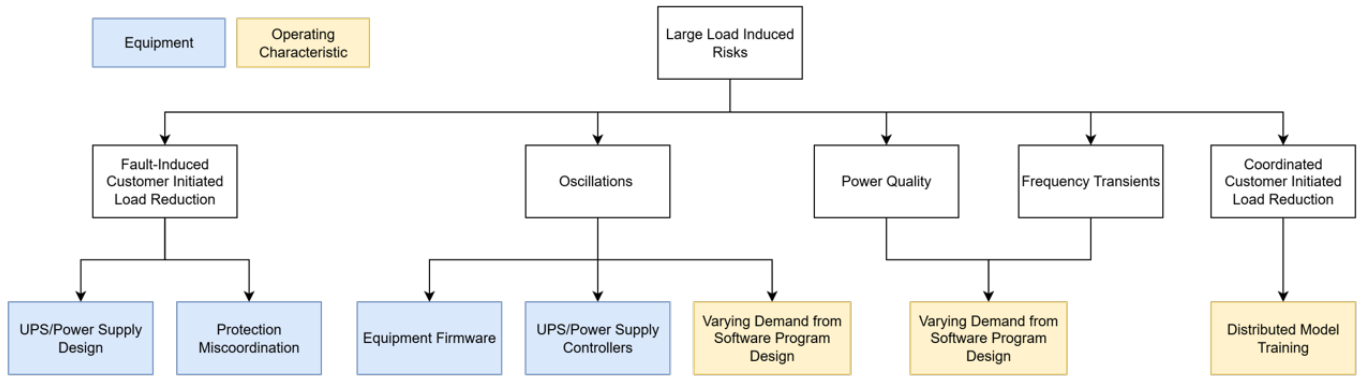


Fig. 1. Taxonomy of the Root Causes of Large Load Induced Bulk Power System Risks

fault [5].

3) *EirGrid*: On the EirGrid transmission system, there were three normally cleared 220 kV transmission system faults (phase-phase fault, single phase to ground fault and reactor fault) in 2022. These faults led to a transient voltage dip that resulted in a reduction of data center demand between 80 MW, 204 MW and 321 MW. Due to strict voltage protection settings (such as a 10% deviation from nominal), the data center loads were moved to UPS within milliseconds of the faults. The power system saw a frequency rise and positive rate of change of frequency (ROCOF) as a result of the significant imbalance triggered by the demand reduction [4].

4) *AEP*: On the AEP transmission system, there have been two normally cleared 138 kV phase-to-ground faults in 2025 within the area of data centers and cryptocurrency mining facilities. These resulted in several facilities tripping after multiple reclosing attempts. Some data centers were identified as having a protection scheme that is set to trip after three voltage disturbances occur within one minute. These data centers rode through the first two reclosing attempts and tripped on the third attempt [13].

B. Oscillations due to Instability in Electronic Controllers

Dominion has seen oscillations due to possible interaction between an unstable UPS controller and a limiter [6]. While the site owner provided the information that the UPS is unstable, no information on what is connected to the site and the possible reasons were shared. The authors in [6] indicated that instability stemming from controllers for power electronic-based devices was responsible for creating oscillations in the range of 14.7 – 14.8 Hz. It was observed that the inclusion of hard limiters in controllers, such as under and over excitation constraints in voltage controllers for equipment protection, may result in chopped state-space once the system trajectory becomes unstable. This results in limit cycles with significantly high oscillation amplitudes.

C. Oscillations due to Outdated Firmware Settings

ERCOT has seen oscillations related to large power electronic loads. On October 25, 2024, a Large Load decreased consumption by 300 MW in 24 seconds. There was no fault

prior to the load consumption decrease; however, there was an oscillation prior to the decrease. The oscillation magnitude was ≈ 25 MW peak to peak, and the oscillation mode was ≈ 23 Hz. Another oscillation was also detected with 20 samples/cycle DFR data showing a ≈ 23 Hz oscillation with ≈ 50 MW peak to peak magnitude on October 28, 2025. At consumption below 300 MW, DFR data (at 20 samples/cycle) showed significant harmonic distortion at the 3rd harmonic but not a 23 Hz oscillation. Following a root cause investigation and testing, the cause was determined to be older firmware versions on certain equipment. The firmware was updated, which resolved the oscillation and improved the harmonic distortion, but was still not in compliance with IEEE 519 [7].

D. Transients due to Regular, Cyclical Fluctuations in Data Center Digital Processes

EirGrid has reported consistent and sustained frequency transients resulting from regular, cyclical fluctuations in data center digital processes. These fluctuations cause frequency jumps between 0.01–0.05 Hz, with minor transients occurring between 10 mHz and 50 mHz at regular intervals have also been observed. These contribute to a significant 5 – 15% drop in steady-state frequency, and interact with frequency response mechanisms including governors and active power controls from renewable energy sources [4].

E. Coordinated Customer Initiated Load Reduction

In AI model training, training jobs can be spread among data centers to distribute the compute load. This is known as distributed model training. If one data center experiences a disruption, then the training is stopped at all the data centers. This could lead to a coordinated customer-initiated load reduction across wide geographic areas. The system operator would have no visibility into this load behavior, thus creating the risk of unplanned load reduction leading to frequency overshoot and overvoltage situations.

F. Root Cause Taxonomy for Large Load-induced BPS Risks

Based on the various real-life events described, a mapping of the root causes of Large Load-induced BPS risks is derived as in Fig. 1. The taxonomy reveals that the root causes are found in both the design of equipment and its controls, along

TABLE III
POTENTIAL SOLUTIONS TO VARIOUS LARGE LOAD ISSUES

Solution	Fault-Induced Load Reduction	Oscillations	Power Quality	Frequency Transients	Voltage Stability
UPS/Power Supply Control Systems Design		✓			
Server Side Load Shaping		✓	✓	✓	
UPS and Power Supply ITIC Curve Change	✓				✓
Protection Coordination	✓				
Grid-Forming Loads	✓				✓
Accurate Dynamic Models	✓	✓			
Improved Monitoring		✓	✓		
Fast Response Ancillary Service	✓				
E-STATCOM	✓	✓	✓	✓	✓
BESS	✓	✓	✓	✓	✓

with the underlying software that the Large Load is operating on. This taxonomy is used to explore both facility-level and grid-level mitigation techniques for the issues identified in the reviewed events. Table III maps the various mitigation techniques to address issues due to Large Load integration, which are discussed in the next two sections.

IV. FACILITY MITIGATION AND DESIGN SOLUTIONS

A. Oscillations, Power Quality, Transients

1) *UPS and Power Supply Control System Design*: Of the known oscillations originating from data centers, all were mitigated with software updates to the control systems of the UPS units/power supply rectifiers, or by changing the code of the underlying software that the Large Load is operating on. Some of the steps include (a) changing the UPS unit input stage control parameters, (b) modifying the power supply rectifier controls, and (c) reducing the loop gain of the UPS rectifier power factor correction circuit.

2) *Server Side Load Shaping*: Other sources of oscillations, power quality issues, and transients can result in varying demand from Large Loads. To mitigate this, software-based solutions that balance computational programs with compute blocks can be implemented. These solutions can monitor the execution of code and the power consumption of servers to optimally schedule jobs executing code in a manner that smooths variations in the server power consumption [8]. Power supply-based solutions to load smoothing could involve adding energy storage to the power supply, software controls to cap the power draw during ramp-ups, and systems to engage processing units with compute jobs to keep the processing unit running to perform a controlled down-ramp.

B. Voltage Ride-Through

1) *UPS and Power Supply Changes*: To improve the low-voltage ride-through capabilities of Large Load facilities, the UPS units and power supplies can be modified. This can be done by modifying the ITIC curve to require equipment to withstand even longer faults and lower voltage depressions. To meet more stringent ITIC requirements, equipment can be upgraded by changing power supply designs to increase the on-board capacitor sizing, or by installing rack-mounted capacitor banks or energy storage.

2) *Coordination between Transmission Protection and Facility Protection Schemes*: As documented in Section II, some of the customer-initiated load reduction occurred due to facility protection schemes transferring the load to backup power after counting three voltage disturbances in a minute. These voltage disturbances originated from normal transmission re-closing practices. Large Load protection schemes may be updated with a timing buffer to account for transmission re-closing practices. Circuit breakers need 40 – 70 ms to operate, and if the event detection timer in the Large Load protection scheme does not account for a breaker timing tolerance, then it may misoperate. If the scheme is set to count a voltage depression at 80 ms with a ± 20 ms tolerance, then it would operate if the voltage depression lasts for 60 ms, which is within the time frame of the breaker operation. Therefore, the protection schemes should be updated to at least 80 – 120 ms to account for breaker operation timing to reduce misoperation rates [12]. This change could potentially prevent data centers from transferring to backup power during events.

To reduce the total count of normal events that can potentially trigger the Large Load transfer to a backup generation, the under-voltage pickup relay settings may be adjusted to ride through system faults. For example, a threshold currently set at 90% of nominal voltage could potentially be relaxed to 85% of nominal voltage while ensuring no damage is caused to low-voltage equipment.

3) *Grid-Forming Loads*: Many Large Loads are inverter-based loads, which can offer additional control capabilities over the load. During a fault, the voltage drops, and the load transfers to backup power and behaves as a passive element. In contrast, Grid-Forming Loads with a voltage source converter interface could be potentially used to regulate and stabilize the voltage at the load following a disturbance in an active manner [14].

V. GRID SIDE MITIGATION AND SOLUTIONS

A. Oscillation and Ride-Through Mitigation

Equipment can be deployed on the grid at the point of interconnection or behind the meter to address Large load-induced oscillations, load variations, and support ride-through capabilities. These include:

1) *E-STATCOM*: These are static compensation devices installed in parallel with the Large Load at or behind the point

of common coupling. They typically have modular multilevel converters that have integrated energy storage systems such as supercapacitors. These supercapacitors have parallel plates with an electrolyte layer in between, and they are designed to charge and discharge in milliseconds. The E-STATCOMs are designed to cycle both active and reactive power (for example ± 75 MW/MVAR for up to several seconds during fault events [15]. This provides a continuous active power supply during the transition of the Large Load to the UPS. Further, reactive power injection during events also helps stabilize system voltage levels. The E-STATCOMs also help smooth alternating load cycles in data centers by injecting or absorbing power into the energy storage unit.

2) *Battery Energy Storage Systems (BESS)*: Similar to supercapacitors, on-site large BESS in parallel to the Large Load can provide quick energy injection and withdrawal capabilities to balance loads. BESS can enable load smoothing during data center training periods where shifts of several tens of MWs can occur several times in a second. While such fluctuations can cause local generation oscillations, generator-generator interaction, inter-area oscillations, and voltage flicker, BESS can reduce load variability up to 70% [16]. BESS can also support low voltage ride-through by mimicking loads just after voltage recovery, drawing power from the grid for charging during the transition of loads from UPS units to the grid.

B. Dynamic Model Improvements

Without accurate dynamic models, many of the documented facility trips and oscillations cannot be replicated [3], [5], [7]. To perform power system planning, dynamic models that accurately represent the load facilities are needed. Large Load developers need to invest in creating dynamic models of their facilities, and the industry should create better load models.

C. Improved Monitoring Capabilities

Both Dominion and ERCOT noted that detecting oscillations and conducting event analysis was challenging due to the lack of high-resolution data [3], [5], [7]. For greater visibility, Large Load developers and transmission owners should deploy high-resolution sensors such as phasor measurement units, digital fault recorders, and power quality monitors at the Large Load point of interconnection. Additionally, this data would help validate dynamic models.

D. Fast Responding Load Ramping Ancillary Service

In ERCOT and EirGrid, the reduction of load following disturbances caused a rise in system frequency. If the system frequency goes too high, generator over-frequency protection will activate, which could worsen system stability. To arrest the frequency rise, a fast responding ancillary service that provides immediate load ramp-up from energy storage resource charging could potentially be implemented.

VI. CONCLUSION

The increasing deployment of Large Loads, such as data centers and crypto-mining facilities, poses numerous reliability

risks to the BPS, as documented in recent transmission system events. This paper categorized the root causes of these risks, which include fault-induced customer load tripping from insufficient Large Load ride-through capabilities and protection system mis-coordination, oscillations from power supply and UPS unit controls and old computing firmware, transients from fluctuations in load demand, and coordinated customer-initiated load reduction from distributed AI model training. To address these risks, this paper provides several mitigation approaches that can be implemented both at the Large Load facility and on the grid side. These include changes to UPS unit and power supply designs, protection coordination, new equipment, grid-forming loads, load shaping, improved monitoring capabilities, and better dynamic models. To this end, further work is needed to address the voltage ride-through capabilities of Large Loads, improve dynamic load models, and to characterize how Large Loads operate to inform the possible risks they pose to the BPS.

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